

6 (a) Define *magnetic flux density*.

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..... [1]

(b) State two situations in which a charged particle in a magnetic field does not experience a magnetic force.

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..... [2]

(c) An electron is travelling at right angles to a uniform magnetic field of flux density 2.0 mT, as shown in Fig. 6.1.

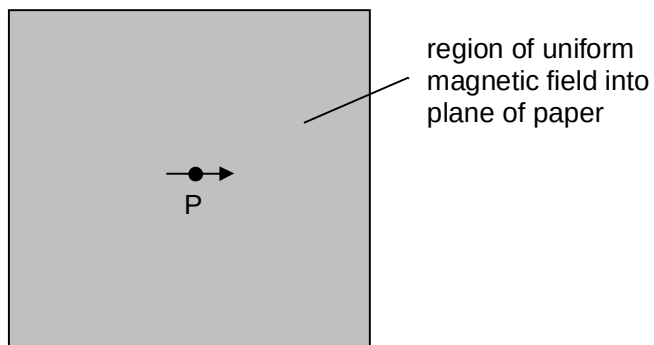


Fig. 6.1

The magnetic field is directed into the plane of the paper. When the electron is at P, its velocity is $2.5 \times 10^7 \text{ ms}^{-1}$, with its direction normal to the magnetic field.

(i) On Fig. 6.1, sketch the path of the electron, assuming that it does not leave the region of the magnetic field. [1]

(ii) Calculate the radius of the electron's path.

radius = m [2]

- (iii) If a uniform electric field is applied in the same direction as the magnetic field, explain the subsequent motion of the electron.

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..... [2]

7 Fig. 7.1 shows an ideal transformer where the primary coil is connected to an alternating